

SWEEP-Q: Smart Wiping & Environmental Exploration Platform – Quadrupe

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I. Abstract

Conventional robotic vacuum systems are effective for routine cleaning on flat and simple surfaces, but their performance is limited when debris is located near obstacles, irregular layouts and hard to reach areas. Most existing household cleaning robots rely on wheeled mobility and fixed vacuum mechanisms, restricting their ability to navigate more complex environments. A quadruped platform with a guided cleaning attachment provides an alternative approach to improving mobility and targeted debris cleaning. This project explores the feasibility of integrating a vacuum-assisted cleaning mechanism into a Unitree A1 robot. The SWEEP-Q prototype uses an OpenManipulator-X robotic arm to guide and stabilize the vacuum nozzle, an Intel RealSense D435i camera to support debris detection, and an onboard Jetson Orin Nano for processing. 3D printed components were also used for modular mounting and hardware integration. The scope of the project is to design, fabricate, and evaluate a proof-of-concept system that demonstrates the integration of quadruped mobility, perception, and guided nozzle positioning.

II. Project Background

SWEEP-Q (Smart Wiping & Environmental Exploration Platform – Quadruped) is a senior capstone design project developed by a team of five Mechanical and Industrial Engineering students at the University of Illinois Chicago (UIC). The project was conducted under the supervision of faculty advisor Dr. Jonathan Komperda and project sponsor Dr. Pranav Bhounsule of the Robotics and Automation Lab, with additional mentorship provided by graduate mentors Chun-Ming Yang, Luca Russo, Paul Peretz, and Joshua Zatz.

The motivation for the project comes from the current capability gap in available robotic household cleaning systems. While commercially available robotic vacuums have become widespread, their effectiveness is highly constrained to flat and unobstructed surfaces. The SWEEP-Q project investigates whether a legged robotic platform, specifically the Unitree A1, can serve as a more capable base for a vacuum-cleaning system. By integrating a robotic manipulator, a depth-sensing camera, and a vacuum unit into the robot, the project aims to demonstrate a proof-of-concept system that improves both mobility and targeted precision cleaning.

III. Problem Statement

Current household cleaning robots are primarily designed for flat, easy-to-navigate environments. They exhibit limited performance in cluttered or uneven indoor spaces. The large majority of commercially available robotic vacuum systems rely on wheeled mobility platforms and fixed vacuum mechanisms. While effective in open spaces, these systems can be hindered by furniture legs, irregular layouts, and surface transitions. These limitations reduce cleaning coverage and frequently require manual human intervention to complete a cleaning cycle. This limitation is particularly significant for users who depend on automation to reduce physical effort, including older individuals and individuals with mobility impairments. The inability of current systems to navigate around obstacles creates a meaningful gap between the capabilities of current available systems and the requirements of a genuinely useful household cleaning robot.

The engineering problem addressed by this project is to design and test a robotic cleaning system that improves the mobility limitation of conventional wheeled platforms available in the market. Additionally, the system must be capable of navigating moderately complex indoor

terrain, positioning a vacuum nozzle with precision, and detecting debris targets using onboard sensing.

IV. Objectives & Engineering Requirements

The primary objective of the SWEEP-Q project is to design, build, and evaluate an autonomous robotic cleaning platform capable of detecting, approaching, and removing small piles of debris in indoor environments. The system integrates a quadruped locomotion base, a robotic manipulator, a depth-sensing camera, and a vacuum unit into a single modular platform. Unlike conventional robotic vacuums, the system is intended to operate without continuous human input and to handle environments that include obstacles, uneven flooring, and debris located in positions that a fixed vacuum system cannot reach.

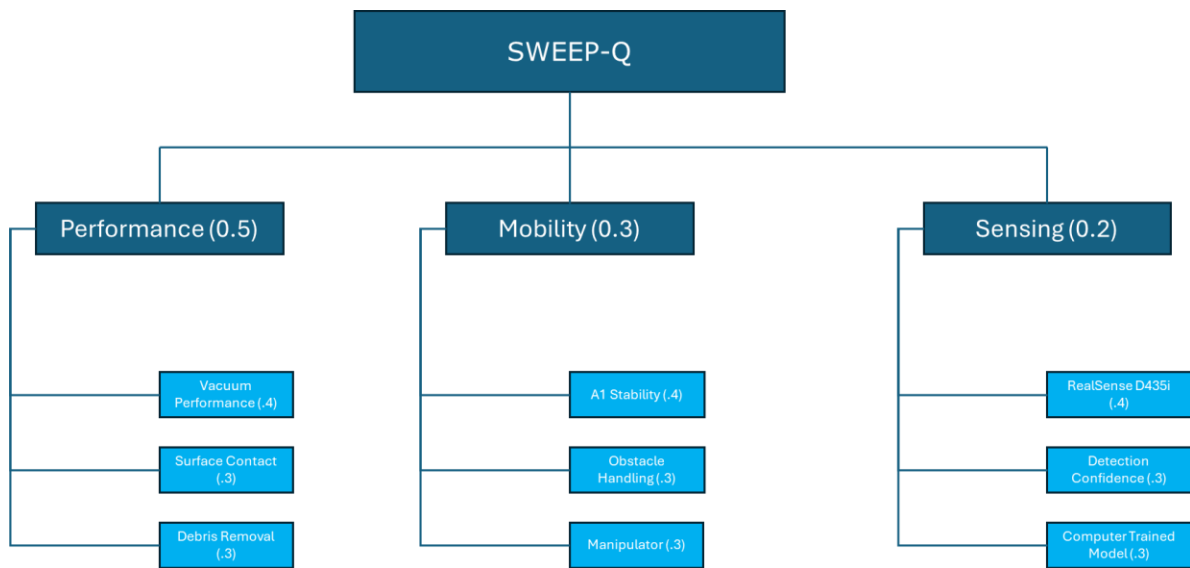


Fig 1. Objective Tree

To translate this objective into proper engineering targets, the team established a set of functional requirements.

The system should be capable of:

- Detecting a target debris pile within the field of view of the onboard camera.
- Estimating the position of the debris relative to its location using depth data.
- Autonomously navigating the quadruped platform toward the target.
- Positioning the vacuum nozzle over the debris using the manipulator.
- Executing a vacuum-cleaning routine to collect the debris.

V. Literature Review

The SWEEP-Q project is informed by prior work in three areas: the architecture of existing autonomous cleaning robots, service-robot deployments in domestic settings, and control of quadruped locomotion platforms.

Şahin and Güvenç [1] survey commercial domestic robots for vacuuming and lawn mowing, including the iRobot Roomba, Karcher RoboCleaner, and Electrolux Trilobite. Their analysis classifies vacuum cleaning robots by chassis shape (simple, round, and arm-equipped) and explicitly notes that adding an arm to a mobile cleaning robot reduces the complexity of obstacle navigation and improves access to corners and narrow spaces. This finding directly motivates the manipulator mechanism adopted in SWEEP-Q, and the review further documents that round-bodied wheeled cleaners cannot effectively reach corners, a limitation that has persisted across the commercial systems that succeeded the ones surveyed.

Service robotics work in domestic contexts supports the use targeted by SWEEP-Q. Gross et al. [2] developed the SERROGA companion robot “Max” for in-home health assistance, integrating navigation, person tracking, haptic interaction, and health-related services such as pulse rate monitoring. Doering et al. [3] complement this with a case study of similar prototype in the apartments of three older adults aged 67-85, identifying narrow passages, variable lighting, and obstacle complexity as the principal challenges for unsupervised in home robot work. Quadruped locomotion control on the Unitree A1, the platform used by SWEEP-Q, has been studied directly by the project sponsor’s research team. Bhounsule et al. [4] demonstrate task-level balance control with Poincare-map-based sim-to-real transfer for high-fidelity command following of the trot gait. Yang and Bhounsule [5] extend this work using Koopman operator theory to enable linear model predictive control of trotting, bounding, and gait transitions on rough terrain. Together these contributions establish that the A1 is a mature platform for the velocity command interface required by an autonomous cleaning routine.

VI. Methodology

a. Design Criteria & Specifications

The team established a set of design criteria based on then project objectives, the capabilities of the available hardware, and the constraints of the budget and academic timeline. The design must accommodate the physical and electrical limitations of the Unitree A1 platform, support modular attachment and removal of all the subsystems, and provide a functional system within an academic year. Cost was a secondary consideration, given the academic context and the use of existing department owned hardware for the most expensive components.

b. Design Alternatives

Three high-level design alternatives were considered for the SWEEP-Q platform. Each alternative reflects a different trade off between mobility, complexity, and targeted cleaning.

Alternative 1 – Wheeled Robotic Base with Fixed Vacuum: The first concept consisted of a traditional wheeled robotic platform equipped with a fixed-position vacuum nozzle. This design prioritizes simplicity, low cost, and ease of control. This design represents the architecture used by most current commercial robotic vacuums. The approach is well suited to flat and open surfaces, but it is fundamentally limited in its ability to navigate uneven terrain, move around obstacles, and direct suction toward specific debris locations. The fixed nozzle position also restricts cleaning flexibility in any application that requires reaching debris located near

furniture, walls, or corners. This approach showed no improvement over existing commercial systems and was hence rejected as a baseline rather than a viable alternative.

Alternative 2 – Quadruped Base with Fixed Vacuum: The second concept replaces the wheeled base with a quadruped robot but retains a fixed position vacuum mounted directly to the robot body. This approach provides substantially improved mobility, allowing the system to traverse obstacles, raised flooring, and uneven surfaces that wheeled robots cannot. However, the absence of a manipulator means that nozzle position is dictated entirely by the robot's body pose, which limits precision in cleaning tasks that require fine positioning, such as collecting debris in tight corners or on the edges of obstacles. The vacuum hose also remains exposed and unsupported, creating the risk of uncontrolled motion during locomotion.

Alternative 3 – Quadruped Base with Manipulator Guided Vacuum: The third concept, and the one selected by the team, adds an active robotic manipulator to the quadruped base. The manipulator stabilizes the vacuum hose during locomotion and actively positions the nozzle over detected debris during the cleaning phase. This approach combines the mobility advantages of the quadruped base with the targeting precision of a cleaning system. The tradeoff is increased system complexity, additional payload, and higher computing demand. However, the added capabilities directly address the engineering gap identifies in the problem statement, and the team determined that the increase in capability justified the additional effort in achieving this build.

Table 1: Design Alternatives Decision Matrix

Design Alternative	Mobility (0.20)	Targeted Nozzle Positioning (0.20)	Expected Cleaning Effectiveness (0.20)	Nozzle / Hose Stability (0.15)	Modularity (0.10)	Cost (0.05)	Ease of Integration (0.05)	Power / Computing Feasibility (0.05)	Total Score
Alt 1: Wheeled Base + Fixed Vacuum	2	1	2	4	2	5	5	5	2.55
Alt 2: Quadruped Base + Fixed Vacuum	5	2	3	2	3	3	3	4	3.10
Alt 3: Quadruped Base + Manipulator	5	5	5	4	4	2	2	2	4.30

Scoring Scale:

1 = Poor, 2 = Fair, 3 = Acceptable, 4 = Good, 5 = Excellent

c. Component Selection

Following the selection of the high-level architecture (Alternative 3), the team conducted formal decision matrix analyses for each major component subsystem: the robotic manipulator,

the handheld vacuum unit, and the onboard computing platform. Each team member independently scored a candidate against weighed criteria after thorough research on all available options. The average scores were used to construct the final decision.

d. Robotic Manipulator

The manipulator subsystem is responsible for stabilizing the vacuum hose during locomotion and positioning the nozzle over detected debris. Two available units appropriate to the project's payload were evaluated: the WidowX 250S and the OpenManipulator-X. The arm must be light enough to fit within the robot's payload (Weight, 0.25), provide sufficient reach for cleaning (Reach, 0.3), maintain a compact form factor to not interfere with locomotion (Size, 0.25), support the limited mass of the hose and nozzle adapter (Payload, 0.10) and remain within budget (Cost, 0.10).

Table 2: Decision Matrix for the robotic manipulator selection.

Arm	Weight	Payload	Size	Reach	Cost	Score
Weighing Factor	0.25	0.1	0.25	0.3	0.1	1
WidowX 250S	4.40	0.00	0.00	6.25	2.86	3.26
OpenManipulator-X	6.92	6.67	0.28	0.00	10.00	3.47

The OpenManipulator-X received the higher overall score and was selected. While the WidowX 250S offers slightly greater reach, the OpenManipulator-X performs better with the Weight and Size criteria and is substantially less expensive. Its reach is sufficient for the cleaning application needed for SWEEP-Q.

e. Battery Powered Vacuum

The vacuum subsystem provides the cleaning function and is required to operate on its own internal battery to keep the cleaning power isolated from the rest of the system. Four candidates were evaluated against the criteria of weight, runtime per charge, debris capacity, size, and cost.

Table 3: Decision Matrix for the vacuum selection.

Vacuum	Supplier	Weight (lb.)	Run Time (mins)	Capacity (qt)	Size	Cost (\$)
7045T48	McMaster-Carr	Null	15	0.5	5.5" x 7" x 13.5"	88.95
7045T56	McMaster-Carr	Null	10	0.5	6.75" x 7" x 18"	148.16

GAS18V-02N (Dremel)	Digikey	2.9	7/Ah	0.5	7.8 " x 17.25 " x 4.7 "	149
CRAFTSMAN Handheld Vacuum Kit	Grainger	4.5	17	0.5	7.625" x 5.25"	193.52

The McCaster-Carr 7045T48 was selected on the combined basis of cost, runtime, and physical size. While the CRAFTSMAN unit offers a slightly longer runtime, its higher cost, and larger footprint disqualified it relative to the project budget and the mounting space available on the A1. The 7045T56 was rejected for its shorter runtime and higher cost.

f. Onboard Computing Platform

Five candidates were evaluated against the criteria of weight, physical size, cost, RAM capacity, and power consumption. The complete decision matrix is shown in Table 4.

Table 4: Decision matrix for the onboard computer.

Board	Weight (.2)	Size (.1)	Cost (.3)	RAM (0.3)	Power (0.1)	Score	Notes
Jetson Orin Nano Dev Kit	2.4	1.5	1	4	1	1.21	8GB RAM, 100×79 mm, ~176 g, 7–25 W, ~\$249
ODROID-N2+ (4GB)	1	1	9.5	1	8.9	2.33	4GB RAM, 90×90 mm, 200 g, 2.3–2.5 W idle / 5.9–6.2 W load, ~\$83
ODROID-M1S (8GB)	9.6	6.8	9.5	4	10	8.16	8GB RAM, 90×65 mm, 52 g, ~3.52 W load, cost \$83 (your value)
Raspberry Pi 5 (16GB)	10	9.4	6.6	10	7.4	11.6	16GB RAM, 86×56 mm, 46 g, ~2.7–3.2 W idle / up to ~9.8 W load, ~\$140
ROCK 4 Model C+ (4GB)	8.6	10	10	1	9.4	4.86	4GB RAM, 85×54 mm, ~70 g (mid of 50–90 g), ~€66 ≈ \$73; approximately 5 W upper typical load

Based on the scoring, the Raspberry Pi 5 (16GB) received the highest overall score (11.60) and was specified as the original computing platform during the planning phase. However, during the prototyping phase, the team determined that the Pi 5 lacked the dedicated GPU acceleration required to run the YOLO computer vision model at a rate sufficient for real time debris identification. The decision was revised mid-project, and the system was moved to the NVIDIA Jetson Orin Nano Developer Kit.

VII. Final System Design

A. Mechanical Design

The mechanical design of the SWEEP-Q system focused on integrating a vacuum-assisted cleaning mechanism onto the Unitree A1 quadruped platform while maintaining system stability, modularity, and mobility. The design process involved development of custom mounting structures, subsystem integration, and iterative improvements to support the robotic arm, vacuum unit, sensing hardware, and onboard electronics. The overall objective of the mechanical subsystem was to create a compact and lightweight structure capable of supporting autonomous cleaning operations in indoor environments.

The system architecture was developed around a manipulator-guided vacuum concept selected through the design alternative evaluation process. Compared to fixed vacuum configurations, the manipulator-guided design provided improved nozzle positioning, better obstacle handling capability, and increased cleaning flexibility in confined areas. The selected configuration required careful consideration of payload limitations, center of mass distribution, and structural support to prevent instability during locomotion and manipulation tasks.

Custom CAD models were created in SolidWorks for subsystem mounting plates, support brackets, vacuum adapters, and arm interface components. Modular design principles were incorporated throughout the system to simplify assembly, maintenance, and future modifications. Several 3D printed components were fabricated to reduce manufacturing cost and allow rapid iteration during testing and prototyping phases. Structural refinements such as rounded corners and fillets were added to reduce stress concentrations and improve durability of printed parts under repeated loading conditions.

One of the primary mechanical challenges involved integrating the robotic manipulator onto the quadruped platform while maintaining sufficient reach to position the vacuum nozzle near ground level. To address this limitation, a custom extension component was designed and fabricated to increase manipulator reach while minimizing added mass. Additional mechanical modifications were also performed on the vacuum housing and hose support structure to improve nozzle stability and reduce unwanted motion during robot movement.

The final mechanical assembly provided stable integration of the vacuum system, robotic arm, camera, and onboard computing hardware onto the Unitree A1 platform. The completed subsystem demonstrated the feasibility of combining quadruped mobility with guided vacuum positioning for autonomous debris cleaning applications in cluttered indoor environments.

B. System Integration Design

The final SWEEP-Q system integrated quadruped locomotion, robotic manipulation, debris detection, and vacuum cleaning into a single autonomous robotic platform. The completed design consisted of five primary subsystems: the Unitree A1 quadruped robot, the

OpenManipulator-X robotic arm, a handheld battery-powered vacuum, the Intel RealSense D435i depth camera, and the NVIDIA Jetson Orin Nano Developer Kit.

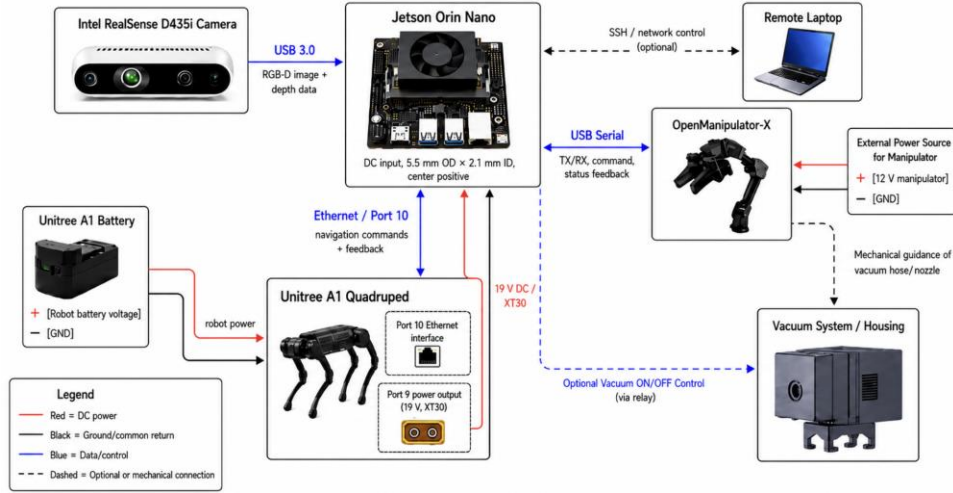


Fig 2. SWEEP-Q hardware wiring and electrical interface diagram.

The SWEEP-Q system integrates the Unitree A1 quadruped, NVIDIA Jetson Orin Nano, Intel RealSense D435i camera, OpenManipulator-X, and vacuum chamber into a single robotic cleaning platform. Figure. 2 shows the high-level hardware wiring and electrical interface for the prototype. The Unitree A1 battery powers the quadruped platform, while the Jetson Orin Nano receives power through the Unitree A1 Port 9 power output. The Jetson serves as the main onboard computer, receiving RGB-D image and depth data from the RealSense camera, sending navigation commands to the Unitree A1 through the Port 8 Ethernet interface, and controlling the OpenManipulator-X through a USB serial connection. The manipulator is powered separately by an external power source and is used to position the vacuum hose and nozzle. The vacuum subsystem remains independently powered, with an optional relay or MOSFET switching circuit shown for future ON/OFF control through the Jetson GPIO.

The Unitree A1 platform served as the primary mobility base and provided the ability to traverse uneven surfaces and navigate around obstacles that conventional wheeled robotic vacuums cannot handle effectively. Mounted onto the platform was the OpenManipulator-X robotic arm, which stabilized and positioned the vacuum nozzle during cleaning operations. The manipulator allowed targeted debris collection by guiding the nozzle directly over detected debris locations.

The cleaning subsystem utilized a battery-powered handheld vacuum selected based on runtime, size, cost, and compatibility with the robot payload limitations. The vacuum system operated independently from the robot power supply to prevent interference with onboard electronics and control systems. A custom nozzle adapter and hose support structure were integrated into the manipulator assembly to improve cleaning precision and reduce hose movement during locomotion.

The sensing subsystem consisted of the Intel RealSense D435i depth camera, which provided visual and depth information for debris detection and target localization. The onboard

computing subsystem used the NVIDIA Jetson Orin Nano Developer Kit to process sensor data and support real-time execution of the YOLOv7 debris detection model. This model was trained with PyTorch on a dataset of about 300 images of sawdust in different angles, shapes, and lighting. This allowed the computer vision system to maintain a locked recognition of the objects of interest, sawdust and powder. The Jetson platform was selected after initial testing demonstrated that alternative computing platforms lacked sufficient GPU acceleration for real-time perception tasks.

Subsystem integration focused heavily on maintaining structural stability, minimizing excess weight, and preserving quadruped mobility. Custom mounting structures and modular 3D printed components allowed efficient integration of mechanical and electrical hardware while simplifying maintenance and future design modifications. Repeated prototyping and testing were performed throughout development to improve subsystem reliability, reduce interference between components, and refine overall system performance.

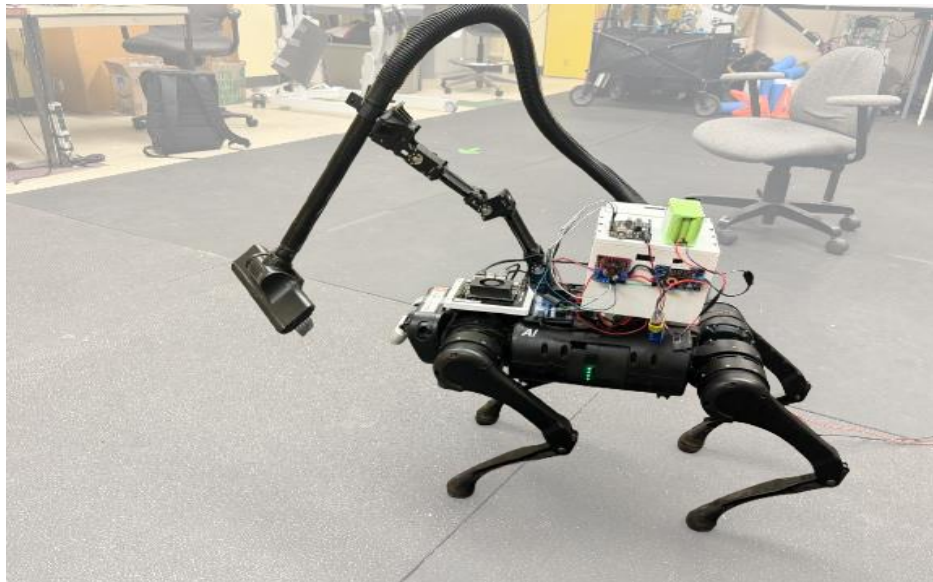


Fig 3: Final assembled SWEEP-Q prototype with integrated quadruped mobility, manipulator-guided vacuum system, perception hardware, and mounting components.

The completed SWEEP-Q prototype successfully demonstrated the integration of autonomous mobility, perception, robotic manipulation, and guided vacuum cleaning within a single robotic platform designed for indoor debris collection tasks.

VIII. System Functional Integration



Fig. 4. System control flowchart for debris detection, navigation, manipulation, and vacuum operation.

The technical architecture of the platform is designed around a unified control sequence that bridges environmental perception with physical execution. The system operates through a continuous feedback loop where raw data from a head mounted RGB-D camera is transformed into actionable motion commands. In the initial phase, a detection algorithm identifies the specific target of sawdust and calculates its spatial coordinates relative to the robot origin. This information is passed to a high level state machine that governs whether the robot should continue its approach or transition into a manipulation phase.

The integration flowchart illustrates a hierarchical progression from data acquisition to mechanical action. The process begins at the perception layer, where the visual sensor captures

environmental data and filters it through a localized detection model. This model outputs specific vectors including the distance and the yaw offset. These variables serve as the primary inputs for the decision layer. The decision logic determines the operational state of the machine based on a proximity threshold.

While the target distance remains above a calibrated 16 inch threshold, the Unitree A1 platform remains in an active locomotion state. During this phase, the integration utilizes proportional control to simultaneously manage forward velocity and angular heading. This allows the robot to home in on the target even when starting from a lateral offset. The internal SDK of the quadruped handles the high frequency stabilization required for a stable gait while the vision system provides the low frequency directional guidance. Once the perception layer confirms that the target is within the designated workspace of the robotic arm, the locomotion system is inhibited to ensure a stable base for manipulation. This handoff is critical to prevent mechanical interference between the moving legs of the dog and the articulating joints of the arm. This ensures that the arm inverse kinematics solver can operate within a predictable and stationary coordinate frame.

IX. Results and Analysis

This section provides a detailed analysis of the performance metrics gathered for the Unitree A1 quadruped platform during target acquisition tasks. The data is categorized by initial starting distance and movement type (longitudinal vs. lateral).

Table 5: Performance at 60-inch Longitudinal Distance

Initial starting distance: 60 in. Objective: Forward approach to target.

Trial	Status	Final Distance (in)	Duration (s)	Observations
1	Fail	N/A	N/A	False positive (detected speck)
2	Success	12	39	Slight chassis tilt
3	Success	15	36	Brief kinematic lag in limb
4	Fail	N/A	N/A	False positive (detected speck)
5	Success	15	35	Slight chassis tilt

Table 6: Performance at 50-inch Longitudinal Distance*Initial starting distance: 50 in. Objective: Forward approach to target.*

Trial	Status	Final Distance (in)	Duration (s)	Observations
1	Success	17	31	Minor chassis tilt
2	Success	17	28	Minor chassis tilt
3	Success	18	29	Minor chassis tilt
4	Success	14	36	Minor chassis tilt
5	Success	15	33	Leftward tilt at terminus

Table 7: Performance for Lateral (Offset) Approaches*Initial base distance: ~50 in. Lateral offset: 10 in.*

Trial	Offset (in)	Status	Final Dist. (in)	Duration (s)	Observations
1	L-10	Success	17	29	Yaw adjusted correctly
2	L-10	Success	17	31	Yaw adjusted correctly
3	L-10	Success	16	29	Yaw adjusted correctly
4	R-10	Success	18	33	Align with target
5	R-10	Success	15	31	Sudden jump in gait
6	R-10	Success	18	31	Yaw adjusted correctly

A. Computer Vision Reliability and Distance Thresholds

The computer vision system demonstrated a significant sensitivity to range. At the 50-inch range and during lateral tests (base 50 in), the system maintained a **100% success rate**. However, when increased to 60 inches, the success rate dropped to **60%**, primarily due to the system incorrectly identifying small surface particulates (specks) as the target. This indicates an "optimal lock-on range" of ≤ 50 in for the current detection parameters.

B. Temporal Efficiency and Lateral Maneuverability

Analysis of the mean durations reveals that Unitree A1 is highly efficient at resolving multi-axis orientation tasks. The lateral offset tests averaged **30.7 seconds**, which is statistically comparable to the straight-line 50-inch trials (**31.4 seconds**). This suggests that the robot's gait controller integrates yaw correction and forward translation simultaneously rather than sequentially, preventing a time penalty for lateral misalignment.

C. Terminal Precision and Kinematic Stability

The platform consistently reached the target within a buffer of 12 to 18 inches. However, a recurring "chassis tilt" was observed in nearly all successful trials. This phenomenon likely stems from the high-frequency corrections issued by the vision-processing stack as the target occupies a larger percentage of the camera's field of view. The recorded instance of a limb malfunction in the 60-inch trial further suggests that extreme ranging may cause computational or mechanical latencies that impact the stabilization algorithm.

Summary

The experimental results validate the Unitree A1's effectiveness for short-range navigation and lateral adjustment. Future work should focus on refining the computer vision noise-filtering to extend reliable detection beyond the 50-inch threshold and improving terminal damping to eliminate the observed chassis tilting.

x. Limitations & Constraints

Several technical and practical constraints limited the final performance of the SWEEP-Q prototype. The most significant limitations involved camera frame rate, limited available documentation for the Unitree A1 software interface, and the team's limited prior experience with robotic system integration.

A. Hardware and Camera Frame Rate

One major limitation was the frame rate of the camera system. The available hardware produced a slower camera feed than desired, which created a delay between target detection and robot response. As a result, the robot sometimes had difficulty maintaining a consistent lock on the sawdust target while moving. In some trials, this delay caused the robot to drift off course or incorrectly identify small surface particulates as the target. A smoother and faster camera feed would likely improve detection reliability and overall navigation accuracy.

B. Unitree A1 Software Integration

Software integration with the Unitree A1 was another major limitation. Compared to more commonly used robotics platforms, the A1 had limited publicly available documentation, shared code examples, and troubleshooting resources. As a result, the team spent significant time understanding the robot's software interface and developing custom code to connect the camera, quadruped platform, and robotic manipulator. This increased development time and limited the team's ability to implement more advanced autonomous features.

C. Limited Prior Robotics Experience

The team's limited prior experience with robotic system integration also affected the project timeline and final performance. Many parts of the project, including hardware integration, manipulator control, computer vision setup, and system-level testing, were new to the team. Because of this learning curve, a significant portion of the project timeline was spent developing basic system functionality before performance optimization could be completed. The final prototype should therefore be viewed as a proof-of-concept system that demonstrates feasibility while leaving room for future refinement.

XI. Design Improvements and Future Work

Although the SWEEP-Q prototype successfully demonstrated the feasibility of combining quadruped mobility, computer vision, manipulation, and vacuum-assisted cleaning, several improvements could be made in future versions of the system. These improvements would focus on making the robot more independent, easier to operate, and more capable of performing different cleaning or service tasks.

One major area for future improvement is the power system. In the current prototype, most of the major components were integrated onto the Unitree A1 platform, but the manipulator still required an external power source to operate. This limited the system's mobility because the robot could not function completely independently from an outlet. A future version of projects should investigate powering the manipulator directly onboard the robot. This could be done using a separate battery pack, a regulated power output from the Unitree A1, or a dedicated onboard power distribution system. The goal would be for the manipulator to operate without any external wiring.

A related improvement would be to make the entire system fully self-contained. Ideally, the Unitree A1, Jetson Orin Nano, camera, manipulator, and vacuum subsystem should all be powered and operated onboard the robot. The vacuum was already designed to use its own internal battery, but future work could improve the overall power integration so that every subsystem is easier to manage and start up. This would allow the robot to operate without any external power sources such as wall outlets, making it more practical for real indoor cleaning applications.

Another possible improvement is the development of a more user-friendly control interface. In the prototype, the system was able to run through a command sent to the Jetson, which allowed the camera, robot, manipulator, and vacuum system to work together. While this was useful for testing, future work could explore creating a mobile or desktop application to control

the system. An app could allow the user to start or stop the cleaning routine, monitor the robot's status, view camera feedback, and select different operating modes without needing to manually access the Jetson through a terminal.

Future versions of the project could also expand SWEEP-Q beyond vacuum cleaning. The Unitree A1 showed that a quadruped platform can support a cleaning attachment while still maintaining useful mobility. Future projects could explore multipurpose cleaning attachments, such as a mop, brush, or wiping tool. The platform could also be adapted for other service tasks, such as light gardening, trimming small shrubs, or carrying inspection tools. A modular attachment system would allow the robot to switch between different tools depending on the task.

Overall, future work should focus on improving onboard power integration, increasing system independence, and expanding the number of tasks the platform can perform. With these improvements, SWEEP-Q could move beyond a proof-of-concept vacuum system and become a more capable multipurpose robotic service platform

XII. Conclusions

The SWEEP-Q project demonstrated the feasibility of integrating quadruped mobility, computer vision, robotic manipulation, and vacuum-assisted cleaning into a single proof-of-concept robotic platform. The final prototype combined the Unitree A1 quadruped, OpenManipulator-X robotic arm, Intel RealSense D435i camera, NVIDIA Jetson Orin Nano, and a battery-powered vacuum subsystem to support targeted debris detection and approach. Testing showed that the system performed most reliably at approximately 50 in, while performance decreased at 60 in due to false detections and vision sensitivity. Overall, the project validated the potential of a manipulator-guided quadruped cleaning system, while also identifying key areas for improvement, including onboard power integration, improved camera performance, filtering, and smoother terminal motion control.

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XIV. Appendices

APPENDIX A. CAD DRAWINGS

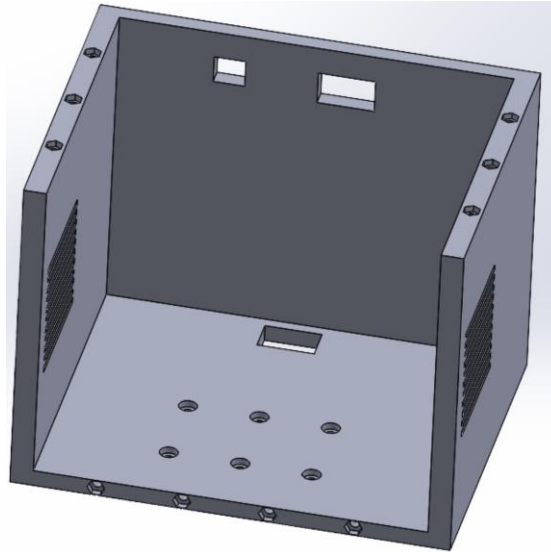


Fig. A1. Internal CAD view of the vacuum housing showing the mounting holes, top cutouts, and side ventilation openings used for component placement and airflow.

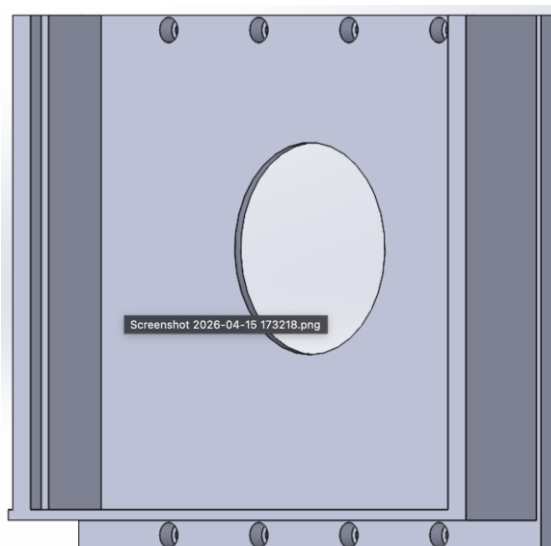


Fig. A2. CAD view of the vacuum housing wall showing the circular nozzle opening and perimeter mounting holes used to attach the wall to the main housing.

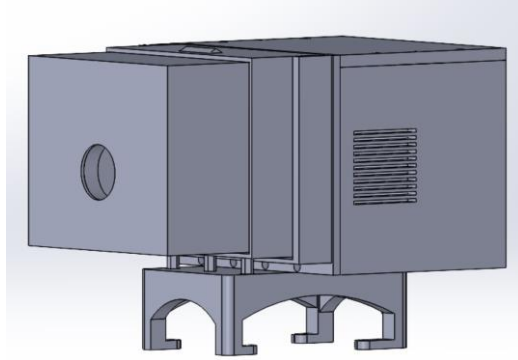


Fig. A3. External CAD view of the assembled vacuum housing showing the front nozzle opening, side ventilation slots, and lower mounting brackets for attachment to the SWEEP-Q platform.